GRAHITH G RAO

AI/ML Engineering Undergraduate · Intelligent Systems · Data Analytics

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PROFESSIONAL SUMMARY

Motivated AI/ML Engineering undergraduate (2023–2027) at AJ Institute of Engineering & Technology with hands-on experience building intelligent systems across NLP, computer vision, and data analytics. Delivered 3 end-to-end projects spanning gesture-recognition hardware, AI-driven career guidance, and fake-review detection. Holder of 4 industry certifications (IBM, AWS, NPTEL). Seeking opportunities to apply data-driven problem-solving and machine learning expertise to create real-world impact in a global environment.

EDUCATION

B.E. — Computer Science & Engineering (AI/ML) · AJ Institute of Engineering & Technology *Aug 2023 – May 2027 (Expected)*

- Relevant coursework: Data Structures & Algorithms, Machine Learning, Deep Learning, Database Systems, Computer Vision, Natural Language Processing.
- Active member of the college AI/ML club; participated in inter-college hackathons and technical fests.

PUC — Science (PCMB) · Canara PU College *2021 – 2022*

PROJECTS

SignSpeak Gloves — Smart Sign Language Converter · Hardware + Software *2024 – Present*

Python · Arduino · TensorFlow Lite · MediaPipe · Android TTS

- Engineered a wearable gesture-recognition glove using flex sensors, MPU-6050 IMU, and an Arduino UNO microcontroller to translate American Sign Language (ASL) gestures into real-time text and audio output.
- Trained a CNN-RNN pipeline (TensorFlow Lite) achieving 92%+ gesture recognition accuracy across all 26 ASL characters, deployable on resource-constrained hardware.
- Designed to bridge communication barriers for an estimated 70 million deaf individuals worldwide; paired with a custom Android TTS application for portable use.

AI-Based Career Guidance System · Software *2024*

Python · Scikit-learn · Random Forest · Streamlit

- Built an end-to-end recommendation engine ingesting academic scores, skill assessments, and interest surveys to suggest optimal career paths via a Random Forest classifier.
- Achieved 88% recommendation accuracy validated against ground-truth career outcomes on a curated dataset; deployed and validated with 50+ college peers via an interactive Streamlit web app.

AI-Based Fake Review Detection System · Software *2024*

Python · NLP · TF-IDF · Logistic Regression · SVM · FastAPI

- Developed a dual-model NLP pipeline (TF-IDF vectorization + Logistic Regression/SVM ensemble) achieving 91% F1-score on a 20,000-review benchmark dataset.
- Engineered behavioral pattern features (review frequency, rating deviation) combined with sentiment analysis to generate per-review trust scores, reducing false-positive rate to under 6%.
- Packaged as a REST API (FastAPI) consumable by third-party e-commerce platforms.

TECHNICAL SKILLS

Languages	Python (primary), SQL, HTML/CSS, JavaScript (basics), C
ML / AI	Scikit-learn, TensorFlow, Keras, NLP (NLTK, spaCy), Computer Vision (OpenCV), Pandas, NumPy, Matplotlib
Data & BI	Data Analysis, Data Visualization, Power BI (basics), Jupyter Notebook, Google Colab
Cloud & Tools	AWS (Cloud Foundations), Git & GitHub, VS Code, Arduino IDE, Streamlit, FastAPI
Design & Other	UI/UX Principles (Figma basics), Canva, Technical Documentation

CERTIFICATIONS

Machine Learning with Python (V2) — IBM / Coursera *2024*
Generative AI Fundamentals — IBM / Coursera *2024*
AWS Cloud Computing Foundations — Amazon Web Services *2025*
Introduction to Machine Learning — NPTEL (IIT) *2023*

LANGUAGES & INTERESTS

Languages: English (Fluent) · Kannada (Native) · Hindi (Conversational)
Interests: AI Ethics & Responsible AI · UI/UX Design · Entrepreneurship